IMEN266 Operations Research II
Chapters 1-3. Basic Probability

Fall, 2026



INDUSTRIAL AND MANAGEMENT
ENGINEERING, POSTECH

Outline

1. Basic probability
2. Basic reliability
3. Bayes' rule
4. Random variable

How to study this chapter (2026 format)

- **Slides** (this file): concepts, class-problem statements, and the worked derivations developed in class. No transcription needed — definitions, theorems and proofs also live in the Companion.
- **Notebooks** — every class problem, twice: in class the `INCLASS` version (fill the `TODO`s, `check()` answers instantly); at home the full version with solutions, sliders, experiments.
- **Companion** (gap-fill $\rightarrow$ full): all definitions and theorems + 6 core proofs with blanks, checkpoints, AI study cards.
- **Homework**: A exam-style (no AI first) · B critique an AI solution · C stress-test an assumption · D prompt log.
- Class rhythm (75 min): quiz $5'$ $\rightarrow$ concepts $\rightarrow$ live notebook $\rightarrow$ wrap-up.
- Ground truth = analytic $\leftrightarrow$ simulation agreement — not AI confidence.
- Materials: PLMS and GitHub `imen266-2026`.

What is it?

- Measure of uncertainty
- Ω: sample space
- Sample point: an outcome of an experiment
- Event: a subset of Ω
- Example) Roll a dice

What is it?

"Measure" is a mathematical term

A probability is a *set function*: it assigns a number to every event (a subset of Ω), with

$$P(\Omega) = 1, \qquad P(\emptyset) = 0.$$

Example: roll a die

- Sample points $1, 2, 3, 4, 5, 6 \quad \Rightarrow \quad \Omega = \{1, 2, 3, 4, 5, 6\}$
- Events: (1) rolling an even number $= \{2, 4, 6\}$; (2) rolling an odd number $= \{1, 3, 5\}$



Properties of events and probability measure

1. $P(\Omega) = 1, P(\emptyset) = 0$
2. If $E_1, E_2, \ldots, E_n$ are mutually exclusive (that is $E_i \cap E_j = \emptyset$ if $i \neq j$),

$$P\left(\bigcup_{i=1}^{n} E_i\right) = \sum_{i=1}^{n} P(E_i).$$

$\Rightarrow$ If $E_1, \ldots$ are mutually exclusive,

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i).$$

3. Let A be an event and A^c its complement (not in A). Then,

$$P(A^c) = 1 - P(A).$$

▷ Take $\lim_{n\to\infty}$ on both sides of the finite sum — the infinite version is *countable additivity*.

Properties of events and probability measure

4. Let A and B be two events,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

5. Inclusion-exclusion law
   If $E_1, \ldots, E_n$ are events,

$$P\left(\bigcup_{i=1}^{n} E_i\right) = \sum_{i=1}^{n} P(E_i) - \sum_{i<j} P\left(E_i \cap E_j\right) + \sum_{i<j<k} P\left(E_i \cap E_j \cap E_k\right) + \cdots + (-1)^n P\left(E_1 \cap \cdots \cap E_n\right).$$

▷ Two-event proof + general idea: Companion, Proof 1.

Conditional probability

- Let A and B be two events. $P(A|B)$ is the probability of A given B.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

- Let A, B and C be three events.

▷ Multiplication rule, typed: Companion §1.2.

Conditional probability

Multiplication rule, and three ways to write the same event

$$P(A \cap B) = P(A \mid B)\, P(B) = P(B \mid A)\, P(A)$$

notation: $P(A \cap B) \;\Leftrightarrow\; P(A, B) \;\Leftrightarrow\; P(AB)$

Three events: conditioning on C

$$P(AB \mid C) = \frac{P(A \cap B \cap C)}{P(C)} = \frac{P(A \mid B \cap C)\; P(B \cap C)}{P(C)}$$

$$= P(A \mid BC)\, P(B \mid C).$$

The highlighted ratio is $P(B \mid C)$: condition everything on C and the multiplication rule reappears.

Independence of events

- Events A and B are independent iff

$$P(A \cap B) = P(A)P(B).$$

$$\Rightarrow P(A|B) = P(A)$$

- Events $E_1, \ldots, E_n$ are (mutually) independent if

$$P(E_1 \cap \cdots \cap E_n) = P(E_1) \cdots P(E_n).$$

- Events $E_1, \ldots, E_n$ are pairwise independent if

$$P(E_i \cap E_j) = P(E_i)P(E_j) \quad \text{for } i \neq j.$$

- Mutually independent $\Leftrightarrow$ pairwise independent?

Independence of events

- Example: Toss a coin twice.

▷ Mutual vs pairwise, typed: Companion §1.2.

Independence of events

Toss a coin twice: pairwise vs. mutual independence

$A \equiv$ both tosses are the same, $B \equiv$ first toss is heads, $C \equiv$ second toss is heads.

$$P(A) = P(B) = P(C) = \frac{1}{2}$$

$$P(A \cap B) = \frac{1}{4} = P(A)P(B), \qquad P(B \cap C) = \frac{1}{4} = P(B)P(C),$$

$$P(C \cap A) = \frac{1}{4} = P(C)P(A) \quad \Rightarrow \text{ pairwise independent}$$

$$P(A \cap B \cap C) = \frac{1}{4} \neq P(A)P(B)P(C) = \frac{1}{8}$$

$\Rightarrow$ **not** mutually independent.

Series system

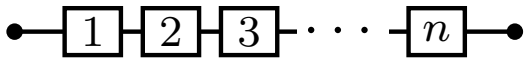
- Consider a system of n components in series.
- Components are all independent.
- p_i: probability that component i functions
- What is the probability that the system functions?

▶ Colab notebook: CP1 warm-up (series/parallel) `[INCLASS] [full]`

Series system

Reliability vocabulary

$X \equiv$ lifetime of the system, $t =$ current time. $P(X > t)$ is the *reliability*; $F(t) = P(X \leq t)$ is the cdf, so the reliability is the complementary cdf $\bar{F}(t) = 1 - F(t)$.



Derivation

$$
\begin{aligned}
P(\text{system works}) &= P(\text{all } n \text{ components work}) \\
&= P(c_1 \text{ works}, \ldots, c_n \text{ works}) \\
&= P(c_1 \text{ works}) \cdots P(c_n \text{ works}) \qquad \text{(independence)} \\
&= p_1 p_2 \cdots p_n \;\leq\; p_i \qquad (i = 1, \ldots, n).
\end{aligned}
$$

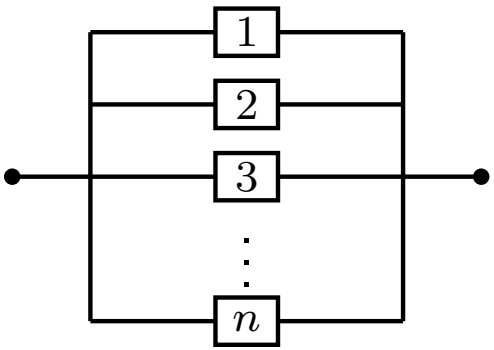
Series systems are weaker than their weakest component.

Parallel system

- Same as series system except the system is composed of n parallel components.
- What is the probability that the system functions?

▶ Colab notebook: CP1 warm-up (series/parallel) `[INCLASS] [full]`

Parallel system



Derivation

$$
\begin{aligned}
P(\text{system works}) &= P(\text{at least one works}) \\
&= 1 - P(\text{all components fail}) \\
&= 1 - P(c_1 \text{ fails}, \ldots, c_n \text{ fails}) \\
&= 1 - P(c_1 \text{ fails}) \cdots P(c_n \text{ fails}) \\
&= 1 - (1-p_1) \cdots (1-p_n) \quad \geq \quad p_i \,.
\end{aligned}
$$

Two one-liners to remember

Parallel systems are stronger than their strongest component; series systems are weaker than their weakest.

▷ "Try": prove the inequality $1 - \prod_j (1-p_j) \;\geq\; p_i$.

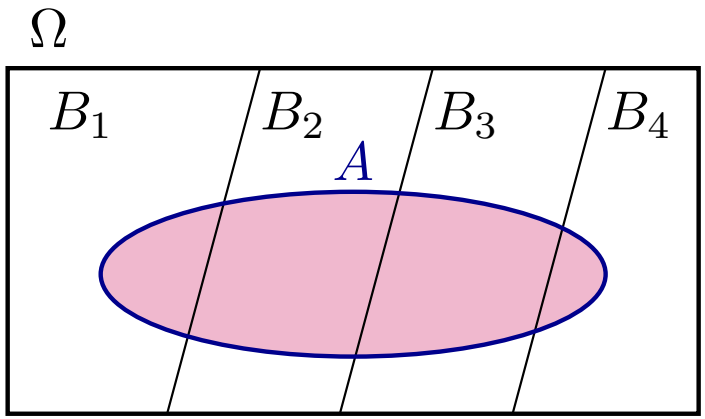
Law of total probability

- Let $B_1, \ldots, B_n$ be n *mutually exclusive* and *exhaustive* events.
  - exhaustive:
  - mutually exclusive:
- Consider an event A in Ω such that
  - $P(A|B_1), \ldots, P(A|B_n)$ are known, and
  - $P(B_1), \ldots, P(B_n)$ are known.
- Law of total probability

$$P(A) = P(A|B_1)P(B_1) + \cdots + P(A|B_n)P(B_n).$$

▷ Statement + proof: Companion, Proof 2.

Law of total probability



The two conditions

mutually exclusive: $B_i \cap B_j = \emptyset$ for $i \neq j$;
exhaustive: $\bigcup_{i=1}^{n} B_i = \Omega$.

Derivation: cut A along the partition

$$\begin{aligned} P(A) &= P(A \cap B_1) + P(A \cap B_2) + \cdots + P(A \cap B_n) \\ &= P(A \mid B_1)P(B_1) + \cdots + P(A \mid B_n)P(B_n). \end{aligned}$$

The shaded pieces $A \cap B_i$ are disjoint because the B_i are; each $P(A \cap B_i) = P(A \mid B_i)P(B_i)$ by the multiplication rule.

Law of total probability

- Four machines produce a total of 10 parts.

	M1	M2	M3	M4
# of parts	2	4	1	3
prob. of defective	0.15	0.05	0.2	0.1

- One part from 10 is randomly selected. What is the probability that it is defective?

Law of total probability

Four machines, ten parts: set up the events first

$A \equiv$ the part is defective; $B_k \equiv$ machine Mk produced the part $(k = 1, \ldots, 4)$.

k	1	2	3	4
$P(A \mid B_k)$ (defect rate)	0.15	0.05	0.2	0.1
$P(B_k)$ (share of parts)	0.2	0.4	0.1	0.3

Then apply the law

$$P(A) = 0.15 \times 0.2 + 0.05 \times 0.4 + 0.2 \times 0.1 + 0.1 \times 0.3$$
$$= 0.03 + 0.02 + 0.02 + 0.03 = 0.10.$$

Class problem 1

1. Three cooks - A, B, and C - bake a special kind of cake, and with respective probabilities 0.02, 0.03, and 0.05 it fails to rise. In the restaurant where they work A bakes 50% of these cakes, B bakes 30%, and C bakes 20%. What fraction of the cakes fail to rise in the restaurant? What is the probability that a given failed cake was baked by cook A?

▶ Colab notebook: CP1 §1 (cooks) `[INCLASS] [full]`

Class problem 1

1. Three cooks: define the events, then it is just LTP + Bayes

$F \equiv$ a cake fails to rise; $A, B, C \equiv$ cook A / B / C baked the cake.

$$P(F \mid A) = 0.02, \quad P(F \mid B) = 0.03, \quad P(F \mid C) = 0.05$$

$$P(A) = 0.5, \quad P(B) = 0.3, \quad P(C) = 0.2$$

$$P(F) = P(F \mid A)P(A) + P(F \mid B)P(B) + P(F \mid C)P(C)$$

$$= 0.010 + 0.009 + 0.010 = 0.029$$

$$P(A \mid F) = \frac{P(A \cap F)}{P(F)} = \frac{P(F \mid A)P(A)}{P(F)} = \frac{0.010}{0.029} \approx 0.345$$

Bayes' rule

- $B_1, \ldots, B_n$ are mutually exclusive and exhaustive given $P(B_1), \ldots, P(B_n)$, $P(A|B_1), \ldots, P(A|B_n)$.
- Bayes' rule

$$P(B_i|A) = \frac{P(A|B_i)P(B_i)}{\sum_{j=1}^{n} P(A|B_j)P(B_j)}.$$

▷ The denominator is $P(A)$ by the law of total probability, so Bayes' rule is just $P(B_i \mid A) = P(A \cap B_i)/P(A)$.

▷ Derivation: Companion, Proof 2 — posterior $\propto$ prior $\times$ likelihood.

Bayes' rule

- Example: COVID TEST
  There is a test which is positive 99% of the time if a person has virus. The test is positive 2% of the time if a person does not have virus. Assume that 1% of population has virus. For a person, if the test is positive, what is the probability that they have virus?

Bayes' rule

Events and data

$A \equiv$ a person has the virus; $T \equiv$ test positive.

$$P(T \mid A) = 0.99, \qquad P(T \mid A^c) = 0.02,$$
$$P(A) = 0.01, \qquad P(A^c) = 0.99.$$

Test vs. reality

	real P	real N
test P	99%	2%
test N	1%	98%

Bayes

$$P(A \mid T) = \frac{P(T \mid A)P(A)}{P(T \mid A)P(A) + P(T \mid A^c)P(A^c)}$$
$$= \frac{0.99 \times 0.01}{0.99 \times 0.01 + 0.02 \times 0.99} = \frac{1}{3}.$$

Class problem 1

2. Suppose that there was a cancer diagnostic test that was 95% accurate on those that do have cancer and 96% accurate on those who don't. If 0.4% of the population have cancer, compute the probability that a tested person has cancer, given that his or her test results indicate so.

▶ Colab notebook: CP1 §2 (cancer test) `[INCLASS] [full]`

Class problem 1

2. Cancer test: the setup

$C \equiv$ a person has cancer; $T \equiv$ test positive.

$P(T \mid C) = 0.95, \qquad P(T^c \mid C^c) = 0.96 = 1 - P(T \mid C^c), \qquad P(C) = 0.004.$

Wanted: $P(C \mid T) = ?$

Finishing it (same route as the COVID example)

$$P(C \mid T) = \frac{P(T \mid C)P(C)}{P(T \mid C)P(C) + P(T \mid C^c)P(C^c)}$$

$$= \frac{0.95 \times 0.004}{0.95 \times 0.004 + 0.04 \times 0.996} = \frac{0.0038}{0.04364} \approx 0.087.$$

Only about 9%: the base rate 0.4% dominates the "95% accurate" test.

Class problem 1

3. Define events appropriately and use Bayes' rule to solve this problem. A gambler has in his pocket a fair coin and a two-headed coin. (a) He selects one of the coins at random, and when he flips it, it shows heads. What is the probability that it is the fair coin? (b) Suppose that he flips the same coin a second time and again it shows heads. Now what is the probability that it is the fair coin? (c) Suppose that he flips the same coin a third time and it shows tails. Now what is the probability that this is the fair coin?

▶ Colab notebook: CP1 §3 (two coins) `[INCLASS] [full]`

Class problem 1

3. Fair coin or two-headed coin: events

$F \equiv$ fair coin, $T \equiv$ two-headed coin, $H_1 \equiv$ first flip is heads, $H_2 \equiv$ first *and* second flips are heads.

(a), (b), (c)

$$\text{(a)} \quad P(F \mid H_1) = \frac{P(H_1 \mid F)P(F)}{P(H_1 \mid F)P(F) + P(H_1 \mid T)P(T)}$$

$$= \frac{\frac{1}{2} \cdot \frac{1}{2}}{\frac{1}{2} \cdot \frac{1}{2} + 1 \cdot \frac{1}{2}} = \frac{1}{3}$$

$$\text{(b)} \quad P(F \mid H_2) = \frac{P(H_2 \mid F)P(F)}{P(H_2 \mid F)P(F) + P(H_2 \mid T)P(T)}$$

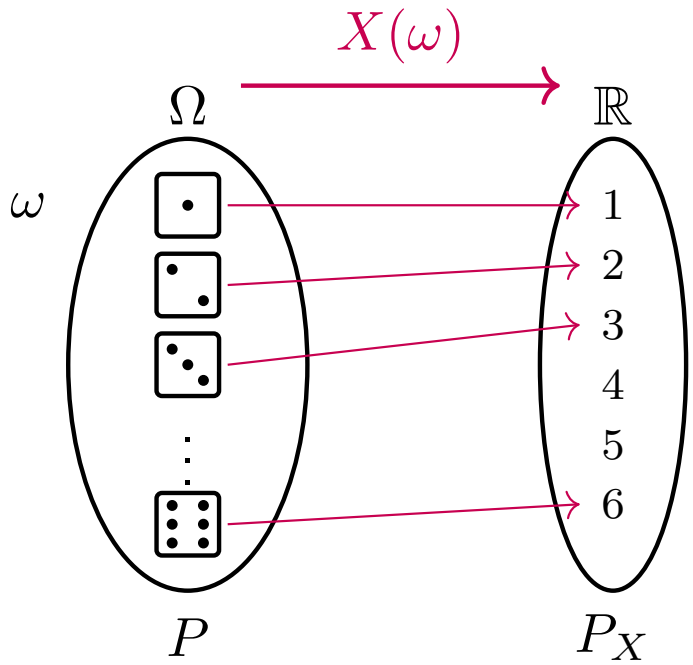
$$= \frac{\frac{1}{4} \cdot \frac{1}{2}}{\frac{1}{4} \cdot \frac{1}{2} + 1 \cdot \frac{1}{2}} = \frac{1}{5}$$

(c) A tail is impossible for the two-headed coin, so $P(F \mid \text{third flip is tails}) = 1$.

Random variable

- A function that maps Ω to $\mathbb{R}$.

Random variable



Roll a die

X counts the dots. Probabilities travel along X:

$$P(\{\text{one, two}\}) = P\big(X^{-1}(\{1,2\})\big) = P_X(\{1,2\}) = P(X \in \{1,2\}).$$

Two kinds, one description

discrete $\to$ pmf (*mass*); continuous $\to$ pdf (*density*). Both give the cdf

$$F(x) = P(X \le x) = P\big(X \in (-\infty, x]\big),$$

and then $P(a < X \le b) = F(b) - F(a)$.

Discrete random variable

- A discrete random variable X is characterized by its probability mass function (pmf) $p(x)$.

$$p(x) = P(X = x).$$

- Example: the probability of producing a defective item is 0.05. Suppose 4 items are produced. Let X denote the number of defectives.

▷ All named distributions, one table: Companion §1.6 (zoo).

Discrete random variable

Properties of a pmf

$$p(x) \geq 0, \qquad \sum_x p(x) = 1.$$

Example: 4 items, defect probability 0.05, X = number of defectives

$$P(X=0) = \binom{4}{0}(0.05)^0(0.95)^4, \qquad P(X=1) = \binom{4}{1}(0.05)^1(0.95)^3, \qquad \ldots$$
$$P(X=4) = \binom{4}{4}(0.05)^4(0.95)^0$$
$$P(X=x) = \binom{4}{x}(0.05)^x(0.95)^{4-x}, \qquad x = 0,1,2,3,4 \qquad \text{(binomial)}.$$

Class problem 1

4. A company has 7 machines on its shop-floor of which 4 are lathes. The service personnel pick two machines at random and check to see if they have any maintenance problems. Let X be the number of lathes selected. Find the probability mass function (or probability distribution) of X. Verify that all properties of PMFs are satisfied.

▶ Colab notebook: CP1 §4 (lathes) `[INCLASS] [full]`

Class problem 1

4. Seven machines, four lathes, pick two: $X \equiv$ number of lathes selected $\in \{0, 1, 2\}$

$$p(0) = P(X = 0) = \frac{\binom{4}{0}\binom{3}{2}}{\binom{7}{2}} = \frac{3}{21}, \qquad p(1) = P(X = 1) = \frac{\binom{4}{1}\binom{3}{1}}{\binom{7}{2}} = \frac{12}{21},$$

$$p(2) = P(X = 2) = \frac{\binom{4}{2}\binom{3}{0}}{\binom{7}{2}} = \frac{6}{21}.$$

Check: all three are ≥ 0 and they add up to $21/21 = 1$. Cdf: $P(X \leq x) = \sum_{k=0}^{\lfloor x \rfloor} p(k)$.

Example distributions

1. Bernoulli random variable
   X is a Bernoulli random variable if it takes value 1 with probability p and 0 with probability $1-p$.

$$p(x) = \begin{cases} p & \text{if } x = 1 \\ 1-p & \text{if } x = 0. \end{cases}$$

2. Binomial random variable
   Consider an experiment comprising n Bernoulli trials. Let X denote the number of 1's in n trials.

$$p(x) = P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}.$$

Example distributions

3. Geometric random variable
   Consider an experiment where Bernoulli trials are performed until a success (or 1) is obtained. Let X denote the number of trials until a success.

$$p(x) = P(X = x) = (1-p)^{x-1}p.$$

4. Negative binomial random variable
   Bernoulli trials are performed until k successes (or 1) are obtained. Let X denote the number of trials until k successes.

$$p(x) = P(X = x) = \binom{x-1}{k-1}(1-p)^{x-k}p^k.$$

Example distributions

Memoryless property of the geometric distribution

Bernoulli trials, $1 =$ success:

trial	1	2	3	4	5	6	7	8
one run	0	0	0	0	0	1		
another	0	0	0	0	0	0	0	1

Stand at trial 3 after three failures: what lies ahead looks exactly like a fresh sequence of trials. The past failures carry no information, so

$$P(X > m + n \mid X > m) = P(X > n).$$

Example distributions

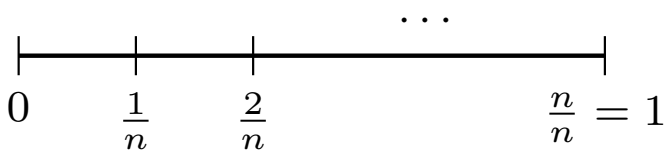
5. Poisson random variable
   X is a Poisson random variable with pmf $p(x)$ if

$$p(x) = P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}, \text{ for } x \in \mathbb{Z} + .$$

Example distributions

Where the Poisson pmf comes from

Split the unit interval into n pieces of length $1/n$, one Bernoulli trial each, and keep the mean fixed: $np = \lambda$. Then let $n \to \infty$.



Binomial $\to$ Poisson

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x} = \frac{n(n-1)\cdots(n-x+1)}{x!} p^x (1-p)^{n-x}$$

$$= \underbrace{\frac{n(n-1)\cdots(n-x+1)}{n^x}}_{\to 1} \cdot \frac{\lambda^x}{x!} \cdot \underbrace{\left(1 - \frac{\lambda}{n}\right)^n}_{\to e^{-\lambda}} \cdot \underbrace{\left(1 - \frac{\lambda}{n}\right)^{-x}}_{\to 1}$$

$$\longrightarrow \frac{e^{-\lambda}\lambda^x}{x!} \qquad (n \to \infty)$$

with $\lim_{x\to\infty}(1 + \frac{1}{x})^x = e$. "Try": check $p(x) \geq 0$, $\sum_x p(x) = 1$.

Class problem 1

5. Historical data has revealed that a manufacturer's current yield is 85%. That is 15% of the parts they produce are defective. If 50 parts are produced.
   (a) What is the probability that exactly 8 parts are defective in total?
   (b) What is the probability that the 10th part produced is the first defective item?
   (c) What is the probability that the 20th part produced is the third defective part?

▶ Colab notebook: CP1 §5 (yield: binom/geom/negbin) `[INCLASS] [full]`

Class problem 1

5. Yield 85%, 50 parts: name the distribution, then write the pmf

(a) exactly 8 defective in total — **binomial**:
$P(8) = \binom{50}{8}(0.15)^8(0.85)^{42}$

(b) the 10th part is the first defective — **geometric**:
$p(10) = (0.85)^9 \times 0.15$

(c) the 20th part is the third defective — **negative binomial**:
$p(20) = \binom{20-1}{3-1}(0.85)^{17}(0.15)^3$

Class problem 1

6. Cars arrive at a certain yield sign on a road according to a Poisson distribution with $\lambda = 8$ cars in a minute.
   (a) What is the probability that exactly 5 cars arrive in a minute?
   (b) What is the probability that more than 2 cars arrive in a minute?

▶ Colab notebook: CP1 §6 (Poisson cars) `[INCLASS] [full]`

Class problem 1

6. Poisson arrivals, $\lambda = 8$ per minute

(a) exactly 5 cars: $P(5) = \dfrac{e^{-8}\,8^5}{5!}$

(b) more than 2 cars: $\displaystyle\sum_{x=3}^{\infty} p(x) = 1 - \sum_{x=0}^{2} p(x)$ (use the complement; never sum an infinite tail by hand)

Continuous random variable

- A continuous random variable X is such that

$$P(X = x) = 0 \text{ for any } x \in \mathbb{R}.$$

- Cumulative distribution function (cdf)

$$F(x) = P(X \leq x).$$

- Probability density function (pdf)

$$f(x) = \frac{d}{dx} F(x).$$

- Properties

▷ cdf properties, typed: Companion §1.5.

Continuous random variable

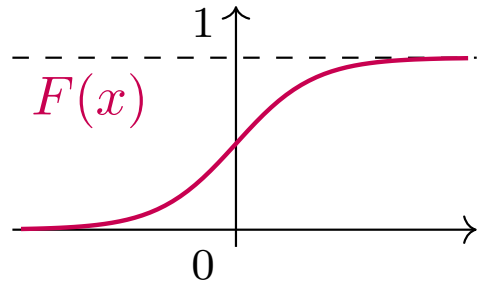
Properties of a pdf and its cdf

$$f(x) \geq 0, \qquad F(x) = \int_{-\infty}^{x} f(u)\,du$$

$$\int_a^b f(x)\,dx = F(b) - F(a) = P(a < X \leq b)$$

$$\int_{-\infty}^{\infty} f(x)\,dx = F(\infty) = 1$$

$$\lim_{x\to\infty} F(x) = 1, \qquad \lim_{x\to-\infty} F(x) = 0$$



Class problem 2

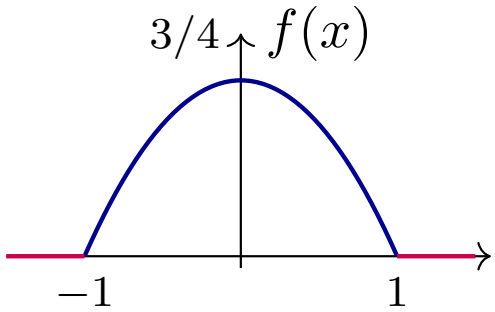
1. Compute $F(x)$, the cdf if X is a continuous r.v. with pdf:

$$f(x) = \begin{cases} \frac{3}{4}(1-x^2) & \text{if } -1 < x < 1, \\ 0 & \text{otherwise.} \end{cases}$$

Also obtain the probability $P(-0.5 < X < 0.75)$.

- Colab notebook: CP2 §1 (polynomial pdf) `[INCLASS] [full]`

Class problem 2



1. Build F piece by piece

$F(x) = \int_{-\infty}^{x} f(u)\,du$, and for $-1 < x \le 1$ this is $\int_{-1}^{x} \frac{3}{4}(1-t^2)\,dt$:

$$F(x) = \begin{cases} 0, & x \le -1, \\ \frac{1}{4}\left(2 + 3x - x^3\right), & -1 < x \le 1, \\ 1, & x > 1. \end{cases}$$

$$P(-0.5 < X < 0.75) = F(0.75) - F(-0.5)$$
$$\approx 0.957 - 0.156 = 0.801.$$

Example distributions

1. Exponential random variable: a random variable X is exponentially distributed with λ if its cdf and pdf are given by

$$F(x) = P(X \leq x) = \begin{cases} 1 - e^{-\lambda x} & \text{if } x > 0 \\ 0 & \text{otherwise.} \end{cases}$$

$$f(x) = \frac{d}{dx} F(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x > 0 \\ 0 & \text{otherwise.} \end{cases}$$

2. Erlang random variable: a random variable X is according to an Erlang distribution with λ and k if its cdf and pdf are

$$F(x) = P(X \leq x) = \begin{cases} 1 - \sum_{r=0}^{k-1} \frac{e^{-\lambda x}(\lambda x)^r}{r!} & \text{if } x > 0 \\ 0 & \text{otherwise.} \end{cases}$$

$$f(x) = \frac{d}{dx} F(x) = \begin{cases} \frac{e^{-\lambda x}(\lambda x)^{k-1}}{(k-1)!} & \text{if } x > 0 \\ 0 & \text{otherwise.} \end{cases}$$

▷ Memorylessness (midterm classic): Companion, Proof 6.

Example distributions

Two remarks on the exponential and Erlang distributions

- Memoryless property of the exponential: $P(X > s + t \mid X > s) = P(X > t)$. "Try to prove it" from $F(x) = 1 - e^{-\lambda x}$.
- Where Erlang comes from: if $X_1, \ldots, X_k \sim \exp(\lambda)$ are independent, then

$$X = \sum_{i=1}^{k} X_i \sim \text{Erlang}(\lambda, k).$$

(e.g. $k = 5$: five exponential stages in a row.)

Example distributions

3. Uniform with parameters a, b

$$f(x) = \frac{d}{dx}F(x) = \begin{cases} \frac{1}{b-a} & \text{if } a \le x \le b \\ 0 & \text{otherwise.} \end{cases}$$

$$F(x) = P(X \le x) = \begin{cases} 0 & \text{if } x < a \\ \frac{x-a}{b-a} & \text{if } a \le x \le b \\ 1 & \text{if } x > b. \end{cases}$$

4. Normal random variable with parameters μ, σ^2

$$f(x) = \frac{d}{dx}F(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < \infty.$$

$$F(x) = P(X \le x) = \int_{-\infty}^{x} f(u)du.$$

▷ All named distributions, one table: Companion §1.6 (zoo).

Class problem 2

2. The lifetime of a single-cell organism is exponentially distributed with parameter 0.1 (per hour). What is the probability that this organism will live for more than 20 hours? What is the probability that it will die within 5 hours?

3. The time for a message to reach www.microsoft.com from POSTECH is distributed according to an Erlang distribution with mean 300 ms and variance 30000 squared-ms. What is the probability that a message will reach within 600 ms? What is the probability it will take more than 900 ms?

▶ Colab notebook: CP2 §2–3 (exponential, Erlang) `[INCLASS] [full]`

Class problem 2

2. Exponential lifetime, $\lambda = 0.1$ per hour; X = lifetime

$$P(X \geq 20) = 1 - P(X < 20) = 1 - F(20) = e^{-2} \approx 0.135$$
$$P(X \leq 5) = F(5) = 1 - e^{-0.5} \approx 0.393$$

3. Erlang: read k and λ off the mean and variance first

$$E[X] = \frac{k}{\lambda} = 300, \qquad \mathrm{Var}(X) = \frac{k}{\lambda^2} = 30{,}000$$
$$\Rightarrow \; k = 3, \; \lambda = 0.01$$
$$P(X \leq 600) = F(600) = 1 - e^{-6}\left(1 + 6 + \frac{6^2}{2}\right) \approx 0.938$$
$$P(X > 900) = 1 - F(900) = e^{-9}\left(1 + 9 + \frac{9^2}{2}\right) \approx 0.006$$

Expectation of a random variable

$$E[X] = \int_{\Omega} X(\omega) dP(\omega)$$

$$= \begin{cases} \sum_x x \cdot p(x) & \text{if } x \text{ is discrete} \\ \int_{\mathbb{R}} x dF(x) = \int_{\mathbb{R}} x \cdot f(x) dx. \end{cases}$$

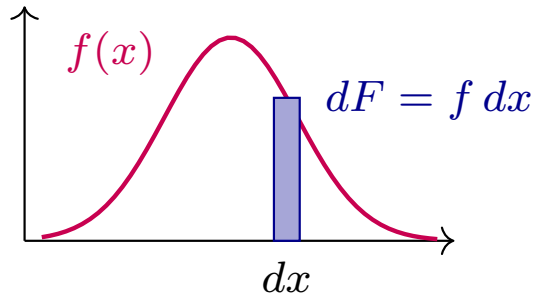
▷ Tail-sum, linearity, LOTUS: Companion, Proofs 3–5.

Expectation of a random variable

One formula for both cases

$$E[X] = \int_\Omega X(\omega)\, dP(\omega) = \int_{\mathbb{R}} x\, dF(x)$$
$$= \sum_x x\, p(x) \qquad \text{(discrete)}$$
$$= \int_{\mathbb{R}} x\, f(x)\, dx \quad \text{(continuous)},$$

because $\dfrac{d}{dx} F(x) = f(x)$, i.e. $dF(x) = f(x)\, dx$.



Expectation of a random variable

- Example: Poisson random variable X with λ.

▷ Result $E[X] = \lambda$: Companion §1.6.

Expectation of a random variable

Poisson mean, worked out

$$E[X] = \sum_{x=0}^{\infty} x\, p(x) = \sum_{x=0}^{\infty} x \cdot \frac{e^{-\lambda}\lambda^x}{x!}$$

$$= \cancel{0 \cdot \frac{e^{-\lambda}\lambda^0}{0!}} + 1 \cdot \frac{e^{-\lambda}\lambda}{1!} + 2 \cdot \frac{e^{-\lambda}\lambda^2}{2!} + \cdots$$

$$= \lambda\Big(\underbrace{e^{-\lambda} + \lambda e^{-\lambda} + \frac{e^{-\lambda}\lambda^2}{2!} + \cdots}_{= \sum_{x\geq 0} e^{-\lambda}\lambda^x/x! \,=\, 1}\Big) = \lambda.$$

Expectation of a random variable

- Example: Exponential random variable X with λ.

▷ One-line route via tail-sum: Companion, Proof 3 (checkpoint).

Expectation of a random variable

Exponential mean, by parts

$$
\begin{aligned}
E[X] &= \int_{\mathbb{R}} x\, dF(x) = \int_{\mathbb{R}} x\, f(x)\, dx = \int_{-\infty}^{\infty} x\, \lambda e^{-\lambda x}\, dx \\
&= \int_0^{\infty} \underbrace{x}_{u}\, \underbrace{\lambda e^{-\lambda x}}_{v'}\, dx \\
&= \Big[-x e^{-\lambda x} \Big]_0^{\infty} + \int_0^{\infty} e^{-\lambda x}\, dx \qquad (u' = 1,\ v = -e^{-\lambda x}) \\
&= 0 + \frac{1}{\lambda} = \frac{1}{\lambda}.
\end{aligned}
$$

Expectation of a function of a random variable

$$E\left[g(X)\right] = \begin{cases} \sum_x g(x) p_X(x) & \text{if } x \text{ is discrete} \\ \int_{\mathbb{R}} g(x) f_X(x) dx & \text{if } x \text{ is continuous.} \end{cases}$$

- Warning: in general

$$E\left[g(X)\right] \neq g\left(E[X]\right).$$

▷ LOTUS proof: Companion, Proof 5.

Expectation of a function of a random variable

Why the formula is legitimate: $Y = g(X)$ is itself a random variable

$$E[g(X)] = E[Y] = \int_{\Omega} Y(\omega)\, dP(\omega) = \begin{cases} \sum_{y} y\, p_Y(y) \\ \int_{\mathbb{R}} y\, f_Y(y)\, dy \end{cases}$$

— the same numbers as $\sum_x g(x) p_X(x)$ or $\int g(x) f_X(x)\, dx$, without ever finding the distribution of Y.

The warning, made concrete with $g(x) = x^2$

$E[X^2] \;\geq\; (E[X])^2$ — the gap is $\mathrm{Var}(X) = E[X^2] - (E[X])^2 \;\geq 0.$

Class problem 2

4. If the cdf of a continuous r.v. is given by

$$F(x) = \begin{cases} 0 & \text{if } x < 0, \\ ax + \frac{bx^3}{3} & \text{if } 0 \le x \le 1, \\ 1 & \text{if } x > 1. \end{cases}$$

If $E[X] = 0.6$, find a and b.

- Colab notebook: CP2 §4 (find a, b) `[INCLASS] [full]`

Class problem 2

5. If X is a continuous r.v. denoting the size of a document at a web server with pdf

$$f(x) = \begin{cases} \frac{24}{x^4} & \text{if } x \geq 2, \\ 0 & \text{otherwise.} \end{cases}$$

Compute the variance of X.

▷ $\mathrm{Var}(X) = E\big[(X - E[X])^2\big] = E[X^2] - (E[X])^2$ — compute $E[X]$ and $E[X^2]$ from the pdf.

▶ Colab notebook: CP2 §5 (variance, heavy tail) `[INCLASS] [full]`